

AI Training SuperServer SYS-820GH-TNR2

8U AI training platform with 8 Gaudi®2 accelerators

Key Applications

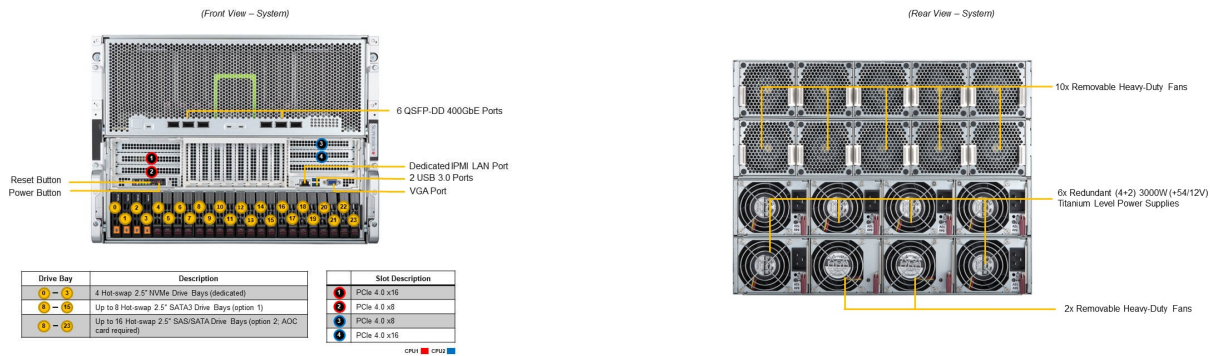
Computer Vision, Natural Language Processing, Recommendation , Purpose Built for AI/Deep Learning Training ,

Key Features

- Dual Intel® Xeon® 3rd Gen Scalable Processors (Ice Lake) TDP up to 270W;
- Intel® C621A Chipset;
- 32x DIMM slots, up to 8TB Registered ECC DDR4 3200MHz SDRAM;
- 2x PCIe 4.0 x16 FHFL slot and 2x PCIe 4.0 x8 FHFL slot;
- 6x 400GbE QSFP-DD for scale-out and 1x 1GbE BaseT (dedicated IPMI port);
- 4x 2.5"NVMe Hot-Swap Drive Bays
Up to 8x 2.5" SATA Hot-Swap Drive Bays
Up to 16x 2.5" SATA Hot-Swap Drive Bays (Additional SAOC is required);
- 12x Removable Heavy-Duty FANs (Includes 2x Dummy PSU FAN);
- 6x 3000W(54V+12V) Redundant Power Supplies, Titanium Level;



Form Factor	8U Rackmount Enclosure: 447 x 356 x 800mm (17.6" x 14" x 31.5") Package: 1020 x 680 x 1400mm (40" x 26.8" x 55.2")
Processor	Dual Socket P+ (LG-4189) 3rd Gen Intel® Xeon® Scalable Processors Up to 40C/80T; Up to 60MB Cache per CPU
GPU	Max GPU Count: Up to 8 double-width GPUs Supported GPU: Intel OAM: Gaudi 2 CPU-GPU Interconnect: PCIe Gen4 x16 GPU-GPU Interconnect: 21x 100GbE PAM4 SerDes Links
System Memory	Slot Count: 32 DIMM slots Max Memory (2DPC): Up to 8TB 3200MT/s ECC DDR4 RDIMM/LRDIMM
Drive Bays Configuration	Default: Total 12 bays • 4 front hot-swap 2.5" NVMe drive bays • 8 front hot-swap 2.5" SATA drive bays Option A: Total 20 bays • 4 front hot-swap 2.5" NVMe drive bays • 16 front hot-swap 2.5" SATA* drive bays (*SATA support may require additional storage controller and/or cables, please see the optional parts list for details) M.2: 2 M.2 NVMe/SATA slots (M-key)
Expansion Slots	Default • 2 PCIe 4.0 x16 FHFL slots • 2 PCIe 4.0 x8 FHFL slots
On-Board Devices	6 QSFP-DD 400GbE 1 RJ45 1GbE
Input / Output	LAN: 1 RJ45 1 GbE LAN port USB: 2 USB 3.0 ports Video: 1 VGA port



System Cooling	Fans: Up to 2 Removable Heavy-Duty 106.5 x 203.5 x 82.4mm Fan(s) Up to 10 Removable Heavy-Duty 80 x 80 x 80mm Fan(s)
Power Supply	6x 3000W Redundant Titanium Level (96%) power supplies
System BIOS	BIOS Type: AMI 32MB SPI Flash EEPROM
Management	SuperCloud Composer®; Supermicro Server Manager (SSM); Supermicro Update Manager (SUM); Supermicro SuperDoctor® 5 (SD5); Super Diagnostics Offline (SDO); Supermicro Thin-Agent Service (TAS); SuperServer Automation Assistant (SAA) New!
PC Health Monitoring	CPU: 7+1 Phase-switching voltage regulator Monitors for CPU Cores, Chipset Voltages, Memory FAN: Fans with tachometer monitoring Pulse Width Modulated (PWM) fan connectors Status monitor for speed control Temperature: Monitoring for CPU and chassis environment Thermal Control for fan connectors
Dimensions and Weight	Weight: Gross Weight: 282 lbs (128 kg) Available Color: Silver
Operating Environment	Operating Temperature: 10°C to 35°C (50°F to 95°F) Non-operating Temperature: -30°C to 60°C (-22°F to 140°F) Operating Relative Humidity: 8% to 80% (non-condensing) Non-operating Relative Humidity: 8% to 90% (non-condensing)
Motherboard	Super X12DPG-OA6-GD2
Chassis	CSE-GP802TS-R000NP